UNIVERSITY OF FREIBURG

Department of Computer Science

Prof. Dr. Josif Grabocka, Prof. Dr. Joschka Boedecker

Machine Learning Course
Winter Semester 2021-2022

Parts	**Points**
1) Linear Methods and SVM	/ 10
2) Bias-variance tradeoff	/ 10
3) Decision Trees and Gradient-Boosted Decision Trees	/ 10
4) Neural Networks	/ 10
5) Ensembles/Boosting	/ 10
6) HPO and Error Measures	/ 10
Total	/ 60

This exam consists of 6 parts worth a total of **60 points** and you have **120 minutes** to answer the questions.

Please check the back of each page, the exam is printed on both sides.

This examination is closed book: **you are not allowed to use any notes**. Please put your name and matriculation number on every page.

Your name: _____ Table number: _____

Your matriculation number: _____

Your signature: _____

Good luck!

Name: _____

Matriculation number: _____

Part 1 – Linear Methods and SVM (10P)

In the toy example of Fig. 1, linearly separable training data points are given.

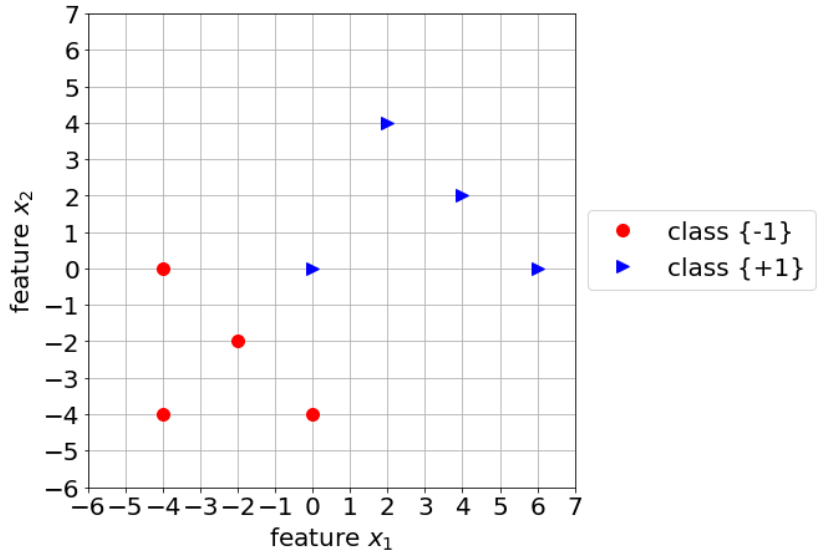


Figure 1: Toy data set #1.

a. **(2P)** Draw the optimal hyperplane to Fig. 1, which will be identified by a linear SVM. Write down the explicit function $f(x_1, x_2) = 0$ of this hyperplane based on the features x_1 and x_2. *Hint:* There is no need for any calculation.

b. **(6P)**

(a) **(3P)** Given that $\|\mathbf{w}\| = 1$ (L2-norm), calculate the values of the weight vector $\mathbf{w}$ and the bias b for the given data in Fig. 1 that the linear SVM optimization will deliver. Briefly describe your approach. What is the dimensionality of $\mathbf{w}$?

(b) **(1P)** Highlight all possible support vectors for the optimal linear SVM solution in Fig. 1. You can either mark them in the figure or write down their coordinates.

(c) **(2P)** Let's assume you know the parameters $\mathbf{w}$ and b of the primal form of a linear SVM. How can you assign the class for an unseen test data point x_{test}? Explicitly give a formula that calculates the label for x_{test}.

Name: _____

Matriculation number: _____

c. **(2P)**

(a) **(1P)** Is vanilla SVM decision hyperplane sensitive to feature scaling? Briefly explain your answer.

(b) **(1P)** Briefly describe an approach so that your SVM could also output the confidence score.

Name: _____

Matriculation number: _____

Part 2 – Bias-Variance Tradeoff and Regularization (10P)

a. **(2P)**

(a) **(1P)** Write (low or high) in the empty spaces below, for the connection between overfitting/underfitting and the Bias-Variance Tradeoff.

Overfitting: _____ bias, _____ variance

Underfitting: _____ bias, _____ variance

(b) **(1P)** Mention briefly two measures (half sentences) to **mitigate overfitting** in:

- Random Forests:

- Linear Regression:

b. **(4P)** Assume that we trained 3 different models on a dataset with three samples $\mathcal{D} = \{x_1, x_2, x_3\}$ using the Mean Absolute Error (MAE) as loss function, and we test it on a sample x_*. The following table shows the prediction of the three models ($f^{(1)}, f^{(2)}, f^{(3)}$) on all the samples.

x	$f^{(1)}$	$f^{(2)}$	$f^{(3)}$	y
x_1	0.1	0	0.2	0.3
x_2	3.9	1	3.5	3.67
x_3	-0.9	2	-0.6	-0.7
x_*	2.0	0.1	0.7	2.3

According to the data in the table above:

(a) **(1P)** Compute the training error and the test error (MAE) for every model (i.e. for each $f^{(1)}, f^{(2)}, f^{(3)}$).

Name: _____

Matriculation number: _____

(b) Answer the following empty spaces with one of the following options: $f^{(1)}, f^{(2)}$, or $f^{(3)}$. In addition, motivate your answers.

- **(1P)** Which model exhibits **better generalization**?: _____

  Motivate:

- **(1P)** Which model exhibits **underfitting**?: _____

  Motivate:

- **(1P)** Which model exhibits **overfitting**?: _____

  Motivate:

c. **(4P)** We train a regression model N times with different training sets. In order to evaluate the model's performance, we compute the predictions on a testing set with M instances. We denote the predicted labels as $f_{i,j}$ and true labels as $y_{i,j}$ respectively, for each trained model indexed by $i \in \{1, ..., N\}$ and test instance indexed by $j \in \{1, ..., M\}$.

Hint: Remember that if you have a set of observations $\{x_1, ..., x_n\}$, the expected value corresponds to the mean, i.e. $\mathbb{E}[x] = \frac{1}{n} \sum_{i=1}^{n} x_i$

Name: _____

Matriculation number: _____

(a) **(1P)** Write the analytic expression for the expected squared error on the testing set. (Use $f_{i,j}$ and $y_{i,j}$ with $i \in \{1, ..., N\}$ and $j \in \{1, ..., M\}$).

(b) **(3P)** Write the analytic expression for the bias and variance of the model on the testing set. (Use $f_{i,j}$ and $y_{i,j}$ with $i \in \{1, ..., N\}$ and $j \in \{1, ..., M\}$)

Name: _____

Matriculation number: _____

Part 3 – Decision Trees and Gradient-Boosted Decision Trees (10P)

a. **(2P)**

(a) **(1P)** What is the **prediction function** of a decision tree with parameters $\theta =< R_1, ..., R_J, \hat{y}^{(R_1)}, ..., \hat{y}^{(R_J)} >$?

(b) **(1P)** For each of the two cases below, simply **draw a decision stump** tree (1 layer with a single decision node), with any decision split condition and leaf weights of your choice.

- Classification Tree:

- Regression Tree:

b. **(4P)** Consider the dataset below with three features and a binary target variable (Default):

Home Owner	**Marital Status**	**Income** $<$ 80**K**	**Default**
Yes	Single	No	No
No	Married	No	No
No	Single	Yes	No
Yes	Married	No	No
No	Single	No	Yes
No	Married	Yes	No
Yes	Single	No	No
No	Single	No	Yes
No	Married	Yes	No
No	Single	No	Yes

- **(3P)** Using the CART algorithm, calculate the Information Gain for all the possible splits for a decision stump tree. Show the decision split with the **largest** Information Gain.

  Provided for ease of calculation:
  $log_2 \frac{7}{10} = -0.51 \quad log_2 \frac{3}{10} = -1.74 \quad log_2 \frac{3}{7} = -1.22 \quad log_2 \frac{4}{7} = -0.81 \quad log_2 \frac{1}{2} = -1$

Name: _____

Matriculation number: _____

- **(1P)** What is the prediction of the decision stump for the test instance below?:

Home Owner	**Marital Status**	**Income** $<$ 80**K**	**Default**
No	Single	No	?

c. **(4P)** Mention one hyperparameter of Decision Trees that requires tuning. Describe the **bias and variance impact** of setting that hyperparameter to a (i) very **low**, or (ii) a very **high** value.

Hyperparameter:

(i) Impact of a very **low** hyperparameter value:

(ii) Impact of a very **high** hyperparameter value:

Name: _____

Matriculation number: _____

Part 4 – Neural Networks (10P)

a. **(2P)**

(a) **(1P)** Mention one advantage and one disadvantage of using a small learning rate in Stochastic Gradient Descent?

(b) **(3P)** I would like to train neural networks for diagnosing the severity of a patient's disease as "stage 1" (least advanced), "stage 2", "stage 3" and "stage 4" (most advanced).

- (1P) How many neurons should I use at the last layer?
- (1P) What activation function should I use at the last layer?
- (1P) What loss function should I use to train the network?

b. **(4P)**

c. **(4P)**

Name: _____

Matriculation number: _____

Part 5 – Ensembles Boosting (10P)

a. **(2P)**

(a) **(1P)** Give one advantage and one disadvantage of using an ensemble of models instead of single individual models.

(b) **(1P)** Which of the following statements are true about Gradient Boosting and Ensembles. Write (**True** or **False**) on the left of the sentence.

i. Gradient Boosting and Ensembles both use multiple models to make predictions.

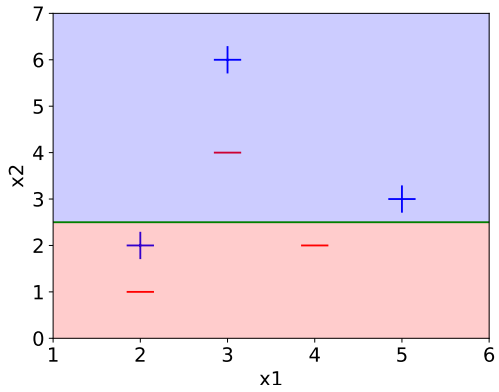
ii. Gradient Boosting always uses one decision stump to make the final prediction.

iii. Gradient Boosting is a sequential algorithm.

iv. Training an Ensemble is a sequential algorithm.

b. **(4P)** AdaBoost. Consider the following dataset in the table at the left. A visualisation of a decision stump tree (1 layer tree with one decision node) is shown in the figure next to the table. The decision split is done on $x_2 > 1.5$.

$\mathbf{x_1}$	$\mathbf{x_2}$	**Class**
2	1	red
3	4	red
3	6	blue
4	1	red
2	2	blue
5	3	blue



Using the **AdaBoost** algorithm, calculate the **error** $Err_1 = \frac{\sum_{i=1}^{6} w_i^{(1)} \cdot \mathbb{I}(y \neq f^{(1)}(x_{i,1}, x_{i,2}))}{\sum_{i=1}^{6} w_i^{(1)}}$, **model weight** α_1 and updated **data point weights** $w_1^{(2)}, ..., w_6^{(2)}$. For the initial weights use $w_1^{(1)} = w_2^{(1)} = ... = w_6^{(1)} = \frac{1}{6}$.

Name: _____

Matriculation number: _____

c. **(4P)** Consider the scenario of fitting an ensemble model on a classification dataset using Gradient-Boosted Decision Trees and Random Forests. Mention briefly (with one or two sentences) **one positive and one negative effect** of **increasing the number of models** for each of these algorithms.

- Gradient-Boosted Decision Trees:

  Positive:

  Negative:

- Random Forests:

  Positive:

  Negative:

Name: _____

Matriculation number: _____

Part 6 – HPO and Error Measures (10P)

a. **(2P)** Briefly explain the difference between Hyperband and BOHB with only few sentences each.

Hyperband:

BOHB:

b. **(4P)**

(a) (3P) Given a binary classifier whose predictions (before threshold) are f for samples with ground truth y, compute and plot the ROC curve (TPR vs. FPR), and compute the AUC. Use the thresholds $\{0, 0.2, 0.4, 0.6, 0.8, 1\}$.

x	f	y
x_1	0.9	1
x_2	0.7	1
x_3	0.5	0
x_4	0.3	1
x_5	0.1	0

Name: _____

Matriculation number: _____

(b) (1P) Which is the best threshold if we want to minimize the number of false positives? Explain your answer.

c. **(4P)** You want to create a house price predictor, which is a machine learning model f that receives a feature vector $\mathbf{x}$ with information about the house (size, location, etc.) and outputs an estimation $\hat{y} = f(\mathbf{x}, \theta)$ of the real price y, where θ are the model parameters.

You want to make the predictions $\hat{y}$ approximate the real price y, but you would like to:

- penalize **more** predictions $\hat{y}$ that fall **below** the **real** price y of the house, and
- penalize **less** predictions $\hat{y}$ that exceed (**above**) the **real** price y of the house.

Design a loss function that helps optimizing θ with respect to the aforementioned desiderata.